

Germain Vivier-Ardisson

PhD Student in Machine Learning and Operations Research

Born: 16/05/2001

Languages: French (native), English (fluent)

Last edited on: August 27, 2026

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POSITIONS AND EDUCATION

Co-Advised PhD , <i>Google DeepMind & CERMICS, ENPC</i>	Paris
Machine learning & operations research. Advised by A. Parmentier and M. Blondel.	Jan 2025 - (Current)
Research Internship , <i>CERMICS, ENPC</i>	Marne-la-Vallée
Combinatorial optimization heuristics as layers. Advised by A. Parmentier and M. Blondel.	Apr 2024 - Aug 2024
Master of Science (M2) , <i>M2A (Learning and Algorithms), Sorbonne Université</i>	Paris
Learning theory, convex and non-convex optimization, generative modeling, GPU programming.	2023 - 2024
Research Internship , <i>MAGI, Polytechnique Montréal</i>	Montréal, Canada
Counterfactual explanations for structured prediction. Advised by T. Vidal.	Mar 2023 - Aug 2023
Research & Development Internship , <i>Quantmetry (now part of Capgemini)</i>	Paris
Benchmark and implementation of hybrid machine learning / operations research algorithms.	Mar 2023 - Aug 2023
Master of Science/Engineering (cycle ingénieur polytechnicien) , <i>Ecole polytechnique</i>	Palaiseau
Applied mathematics, computer science, physics.	2020 - 2024
Bachelor of Science (classes préparatoires PTSI/PT) , <i>Lycée Privé Sainte-Geneviève</i>	Versailles
Mathematics, physics, engineering. Ranked 4th at Ecole polytechnique's entrance exam.	2018 - 2020

TEACHING EXPERIENCE

- **Teaching Assistant**, Introduction to convex optimization and linear programming at ENPC (1A) Fall 2024
- **Oral Examiner in Mathematics** at Lycée Saint-Louis (Colleur en filière PSI) Sep 2023 - Jun 2024

PUBLICATIONS

- [1] **G. Vivier-Ardisson**, L. Demonet, A. Parmentier, and M. Blondel, “[Regularized Large Neighborhood Search](#)”, *arXiv preprint arXiv:2606.02294*, 2026.
- [2] **G. Vivier-Ardisson**, M. E. Sander, A. Parmentier, and M. Blondel, “[Differentiable Knapsack and Top-k Operators via Dynamic Programming](#)”, *arXiv preprint arXiv:2601.21775*, 2026.
- [3] M. Blondel, M. E. Sander, **G. Vivier-Ardisson**, T. Liu, and V. Roulet, “[Autoregressive Language Models are Secretly Energy-Based Models: Insights into the Lookahead Capabilities of Next-Token Prediction](#)”, *International Conference on Machine Learning*, 2026.
- [4] **G. Vivier-Ardisson**, M. Blondel, and A. Parmentier, “[Learning with Local Search MCMC Layers](#)”, *Transactions on Machine Learning Research (Featured & J2C Certifications)*, 2026.
- [5] **G. Vivier-Ardisson**, A. Forel, A. Parmentier, and T. Vidal, “[CF-OPT: Counterfactual Explanations for Structured Prediction](#)”, *International Conference on Machine Learning*, 2024.